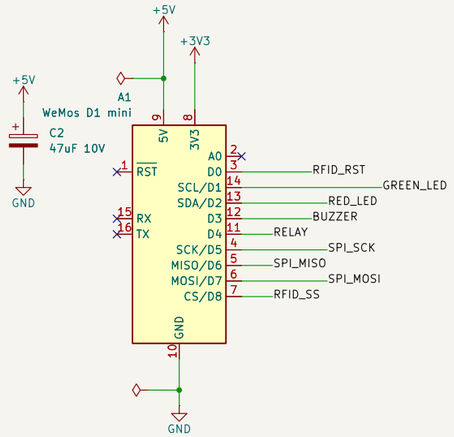
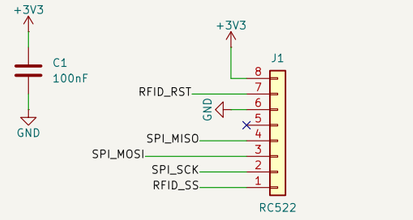
WEMOS D1 MINI (ESP8266) -- USB POWER



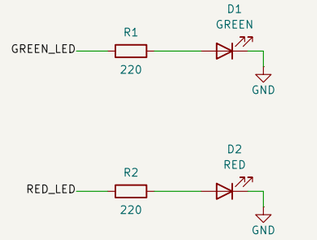
Plugs into two 1x8 female headers. Powered from its micro-USB socket (5 V, >= 500 mA). Total draw < 400 mA.
Boot straps: D3/GPIO0 and D4/GPIO2 HIGH, D8/GPIO15 LOW at reset.
The 47k base resistor leaves D3 at ~2.8 V (VIH 2.5 V).
C2: bulk for buzzer and relay-module switching.

RC522 RFID READER



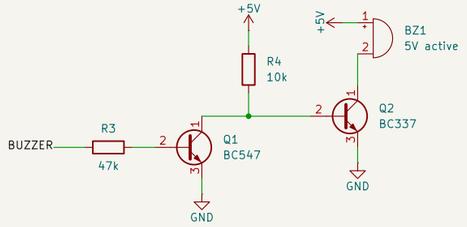
Module pin order: SDA SCK MOSI MISO IRQ GND RST 3.3V.
3.3 V ONLY -- from the D1 mini's regulator (~26 mA).
IRQ unused. Plug the module in, or run a cable <= 15 cm.

STATUS LEDS



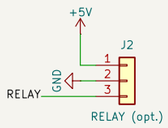
Active HIGH. (3.3 V - 2.0 V) / 220 = ~6 mA.
5 mm LEDs: long leg (anode) toward the resistor.

BUZZER DRIVER (silent at boot)



D3 HIGH (boot, idle): Q1 on, Q2 off, silent.
D3 LOW: buzzer on. Firmware BUZZER_ACTIVE_LOW true.
BZ1: 5 V ACTIVE buzzer (beeps on DC). + pin to +5V.

OPTIONAL RELAY MODULE



1 VCC 2 GND 3 IN, for an active-LOW 5 V relay module (RELAY_ACTIVE_LOW true). D4 is HIGH at boot: relay off. Give the lock its OWN supply -- never the USB 5 V -- and fit a flyback diode across the lock coil.

Wemos D1 mini + RC522 carrier, USB powered

card-reader

Sheet: /

File: rfid-door-controller.kicad_sch

Title: RFID Attendance Device

Size: A3	Date: 2026-11-03	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 1/1